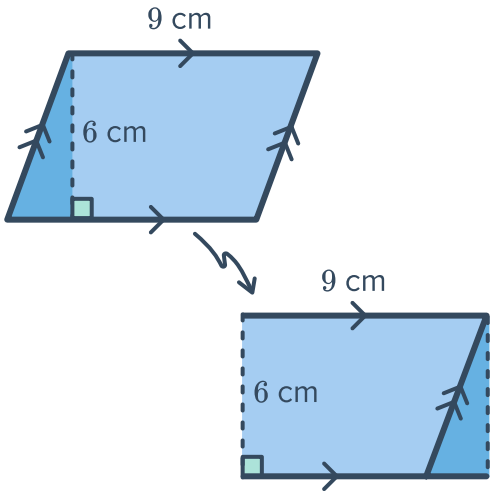
Parallelograms

1 The given parallelogram is formed into a rectangle:

a Find the area of the rectangle.

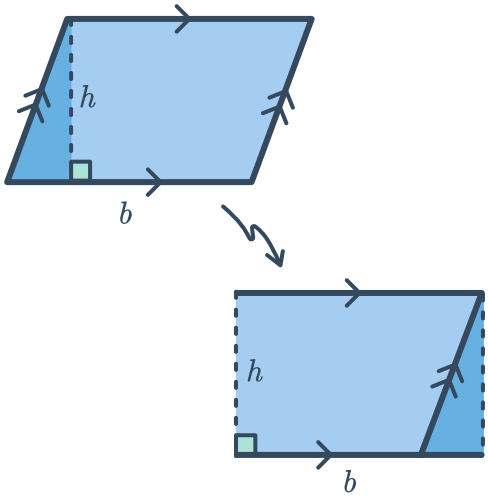
b Hence, find the area of the parallelogram.



2 The given parallelogram is formed into a rectangle:

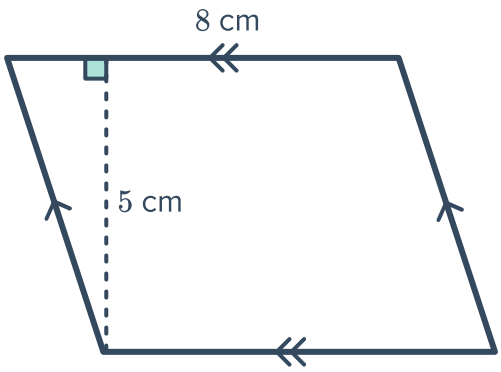
a Find an expression for the area of the rectangle in terms of b and h.

b Hence, find an expression for the area of the parallelogram in terms of b and h.

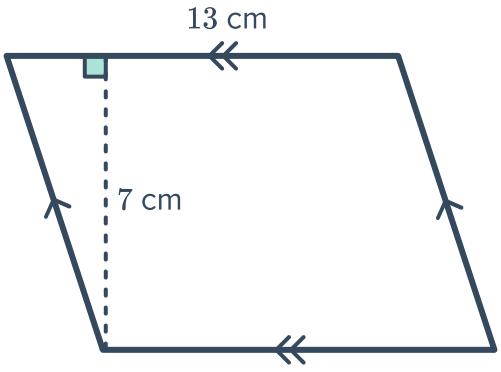


3 Find the area of the following parallelograms:

a

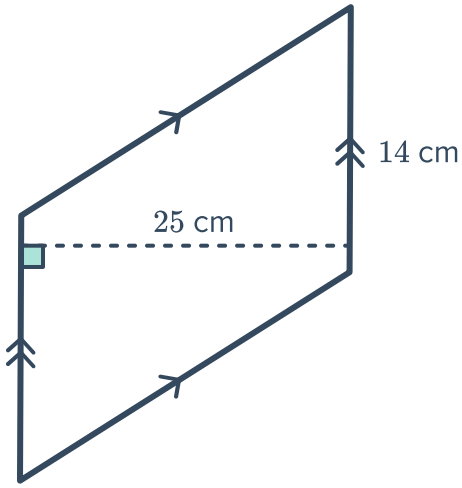


b

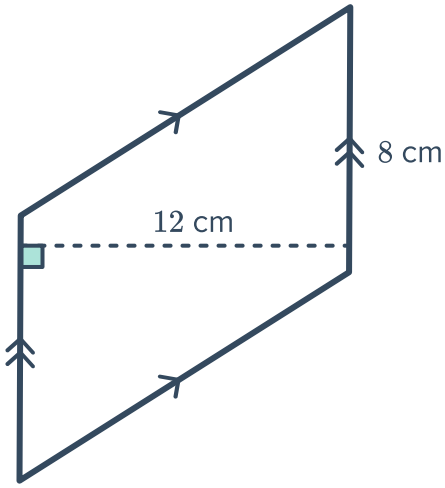


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c



d



4 Determine whether the following pairs of values could be the dimensions of a parallelogram with an area of $70\,\text{mm}^2$.

a Base $= 10\,\text{mm}$, Height $= 7\,\text{mm}$

b Base $= 7\,\text{mm}$, Height $= 10\,\text{mm}$

c Base $= 1\,\text{mm}$, Height $= 70\,\text{mm}$

d Base $= 2\,\text{mm}$, Height $= 70\,\text{mm}$

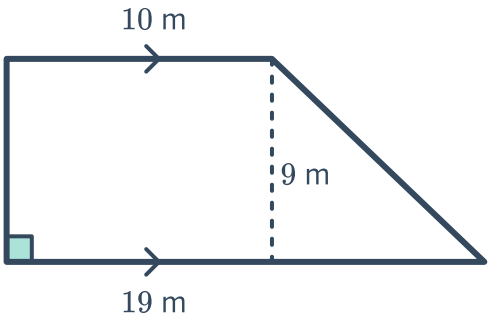
Trapeziums

5 The given trapezium is split into a rectangle and a right-angled triangle:

a Find the area of the rectangle.

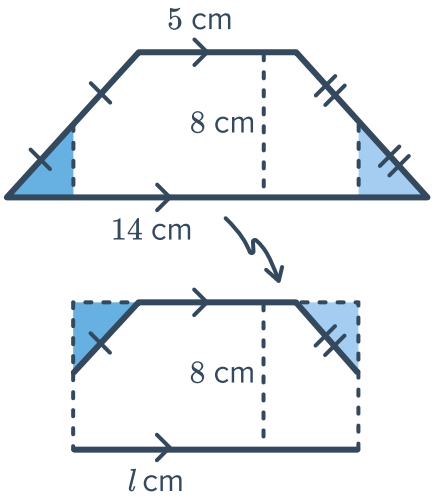
b Find the area of the triangle.

c Hence, find the area of the trapezium.



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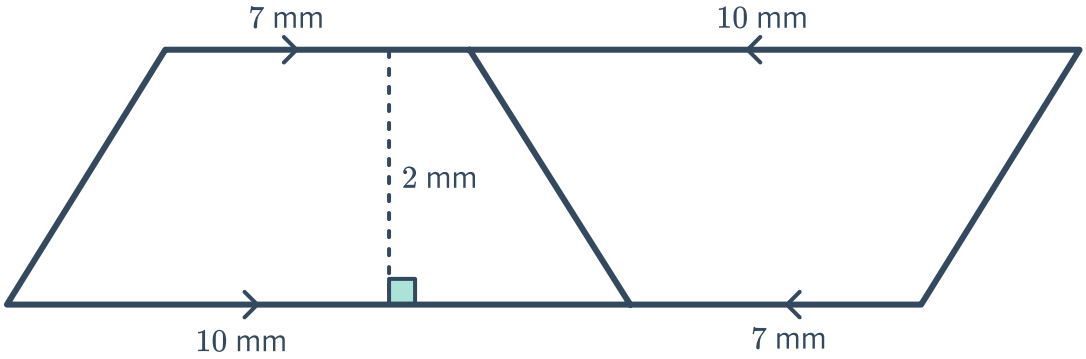
6 The given trapezium is formed into a rectangle:



a Find the length, l, of the rectangle.

b Hence, find the area of the trapezium.

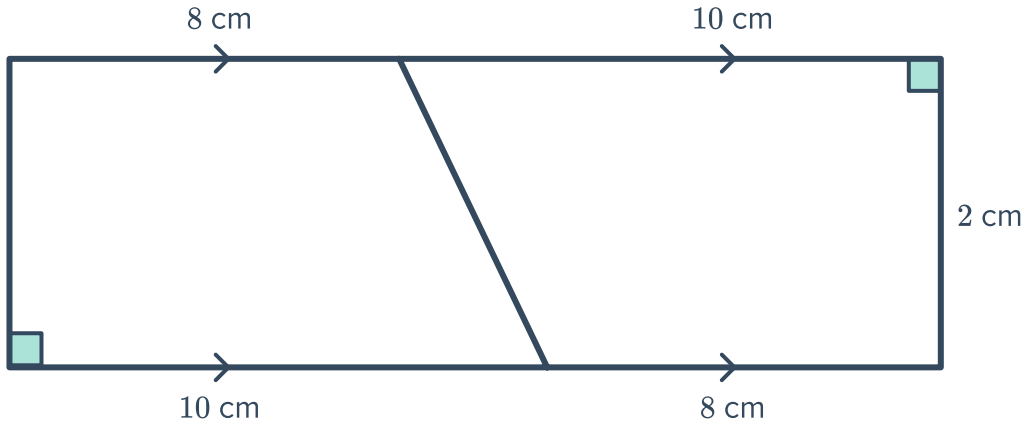
7 Two identical trapezia are put together to make a parallelogram:



a Find the area of the entire parallelogram.

b Find the area of one of the trapezia.

8 Two identical trapezia are put together to make a rectangle:

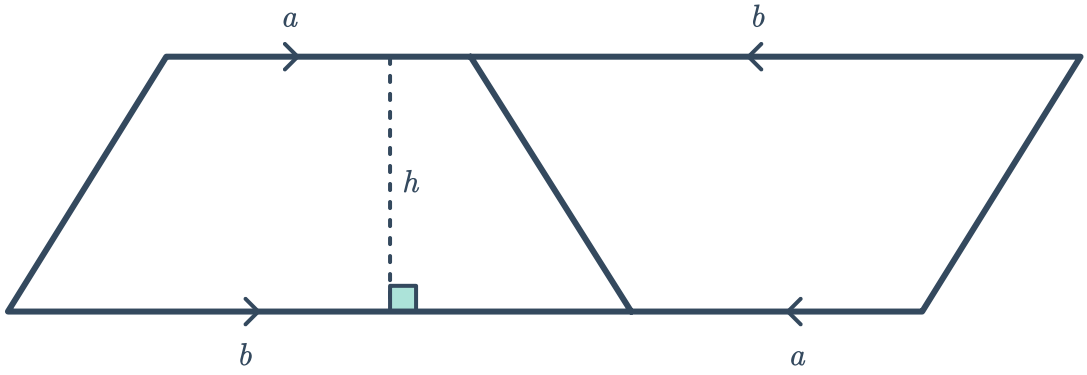


a Find the area of the entire rectangle.

b Find the area of one of the trapezia.

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9 Two identical trapezia are put together to make a parallelogram:

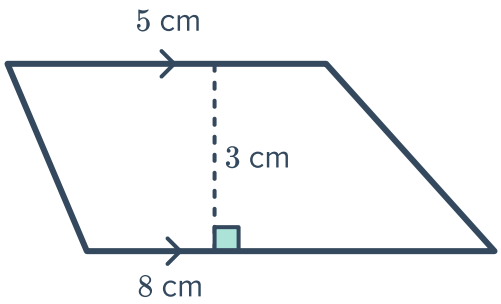


a Find an expression for the area of the entire parallelogram in terms of a, b and h.

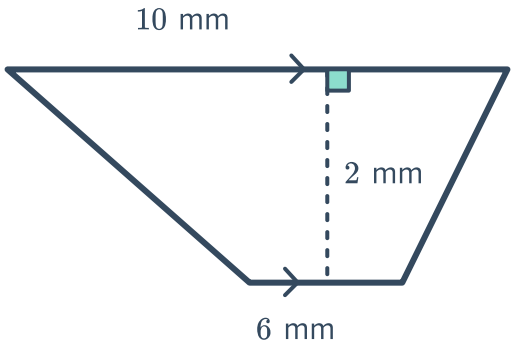
b Find an expression for the area of one trapezia in terms of a, b and h.

10 Find the area of the following trapeziums:

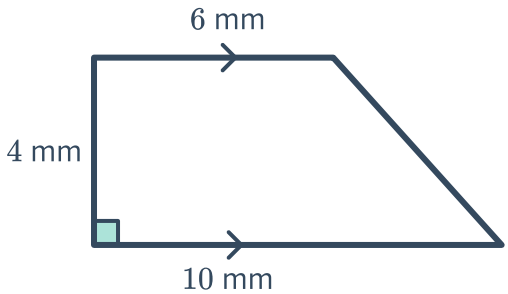
a



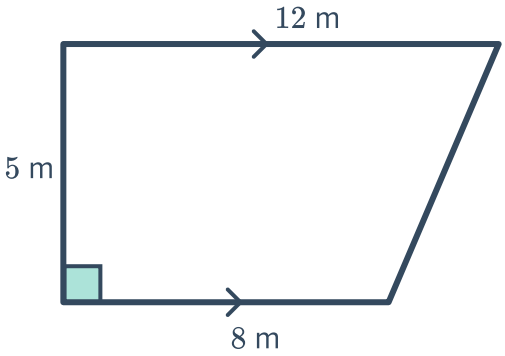
b



c

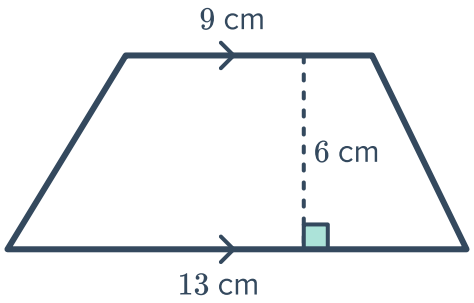


d

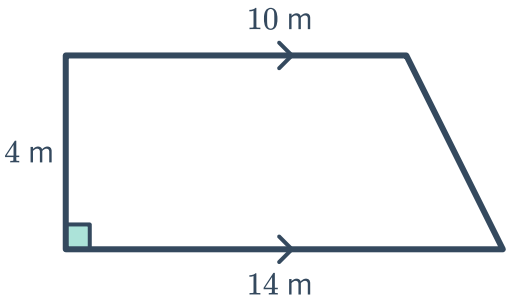


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e



f

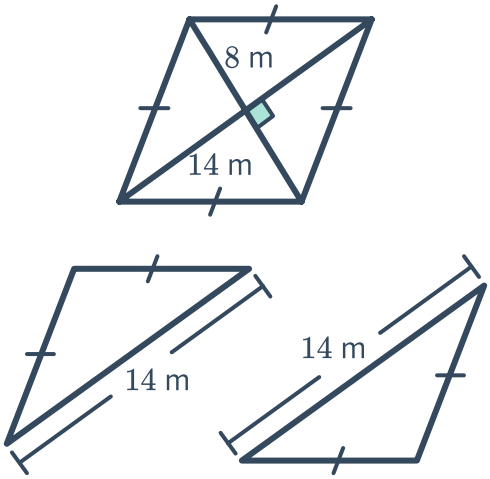


Rhombuses

11 The given rhombus can be split into two triangles:

a Find the area of one triangle.

b Hence, find the area of the rhombus.



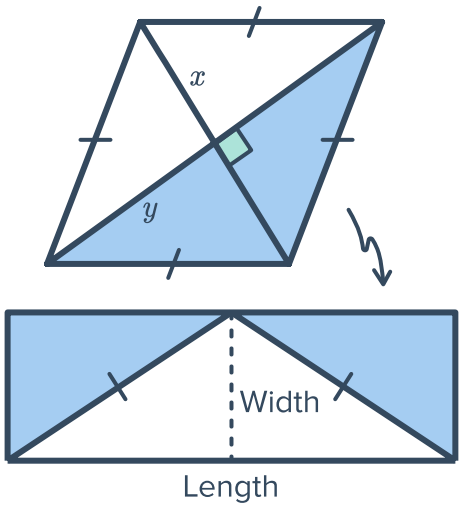
12 The given rhombus is formed into a rectangle:

a Find the length of the rectangle in terms of y.

b Find the width of the rectangle in terms of x.

c Find the area of the rectangle in terms of x and y.

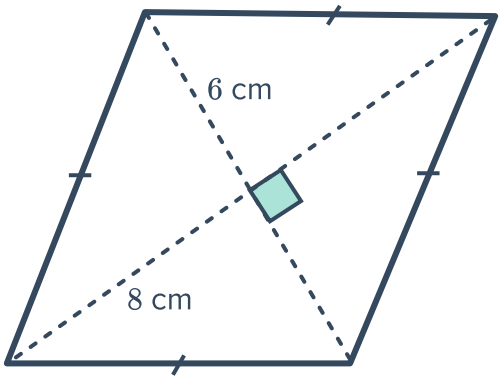
d Hence, find the area of the rhombus in terms of x and y.



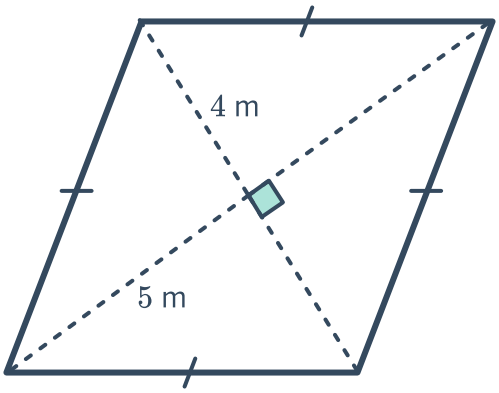
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13 Find the area of the following rhombuses:

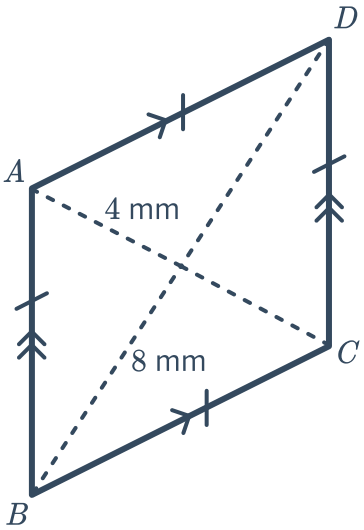
a



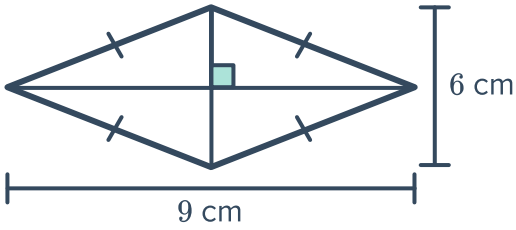
b



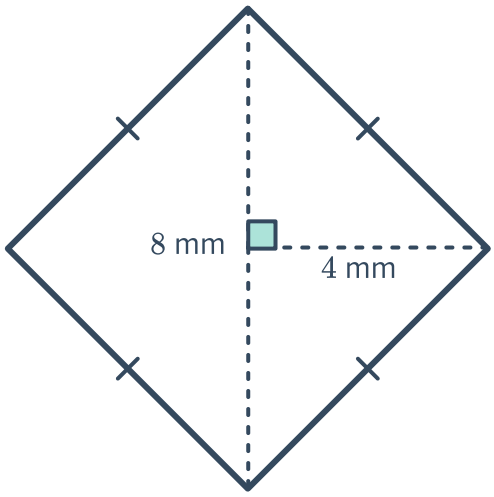
c



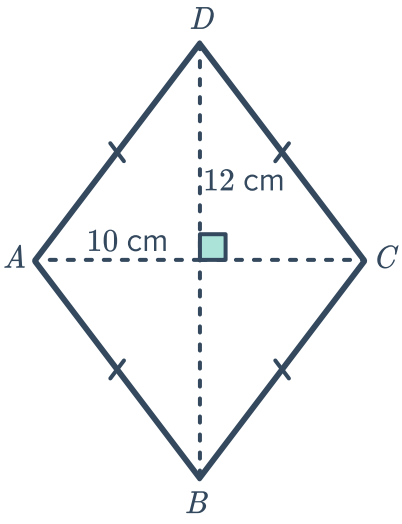
d



e



f



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14 Determine whether the following pairs of values could be the diagonal lengths, x and y of a rhombus with an area of $9\,\text{m}^2$.

a $x = 2\,\text{m}$ and $y = 9\,\text{m}$.

b $x = 6\,\text{m}$ and $y = 3\,\text{m}$.

c $x = 12\,\text{m}$ and $y = 3\,\text{m}$.

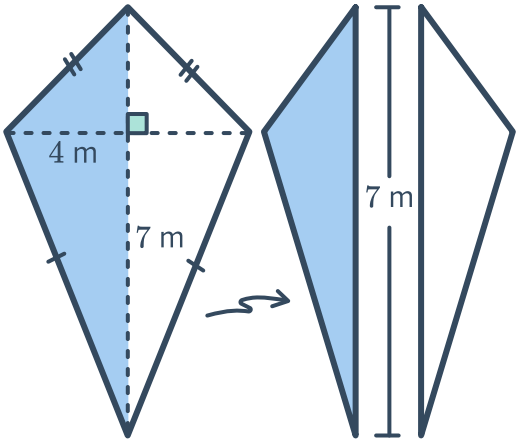
d $x = 6\,\text{m}$ and $y = 6\,\text{m}$.

Kites

15 The given kite can be split into two triangles:

a Find the area of one of the triangles.

b Hence, find the area of the kite.

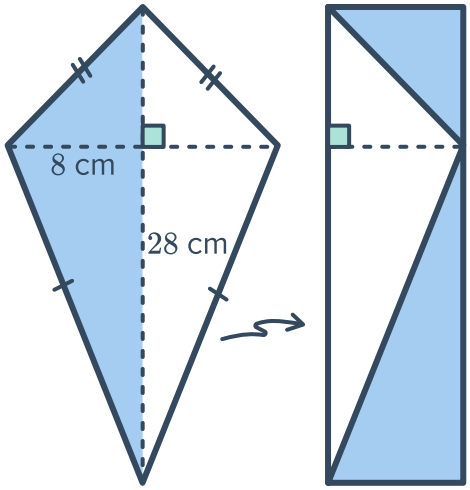


16 The given kite is formed into a rectangle:

a Find the length of the rectangle.

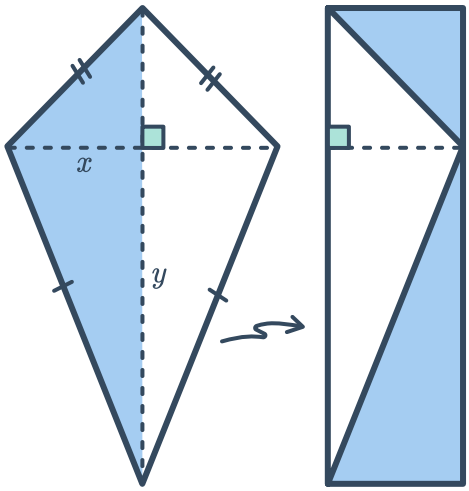
b Find the width of the rectangle.

c Hence, find the area of the kite.



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17 The given kite is formed into a rectangle:



a Find the length of the rectangle in terms of y.

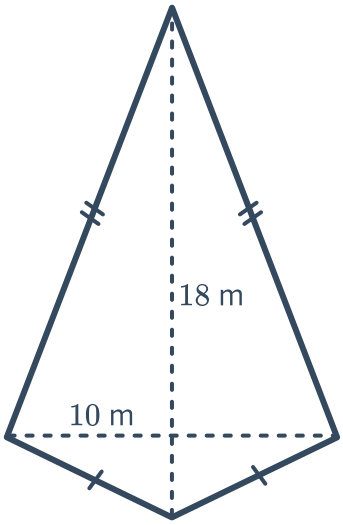
b Find the width of the rectangle in terms of x.

c Find the area of the rectangle in terms of x and y.

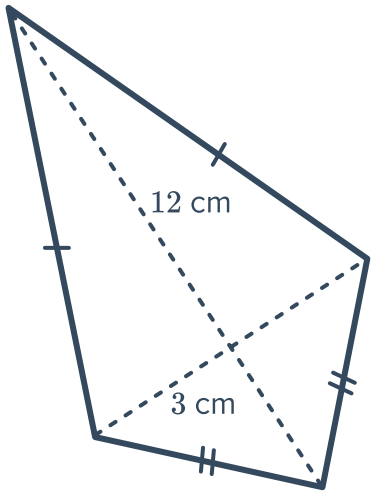
d Hence, find the area of the kite in terms of x and y.

18 Find the area of the following kites:

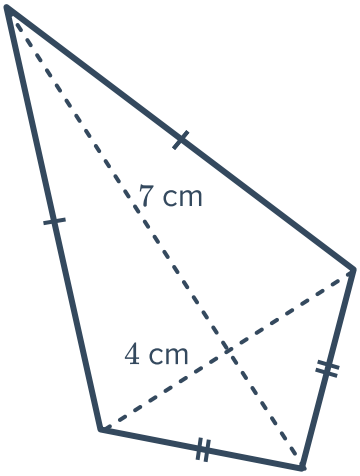
a



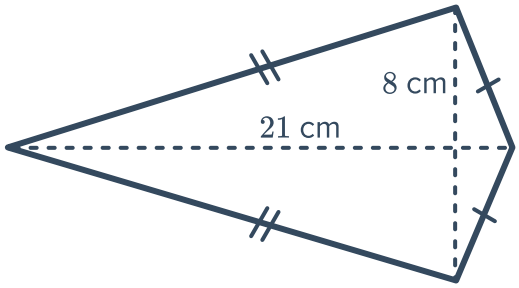
b



c

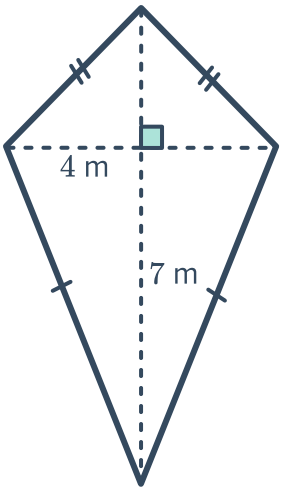


d

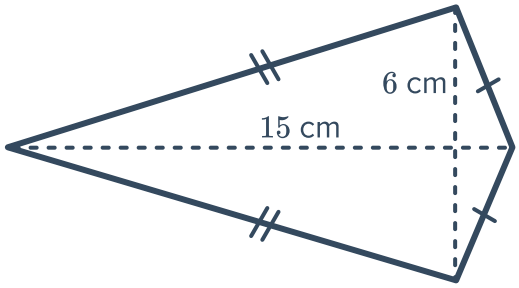


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e



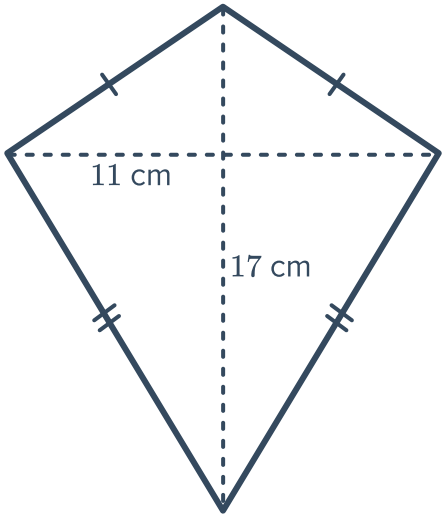
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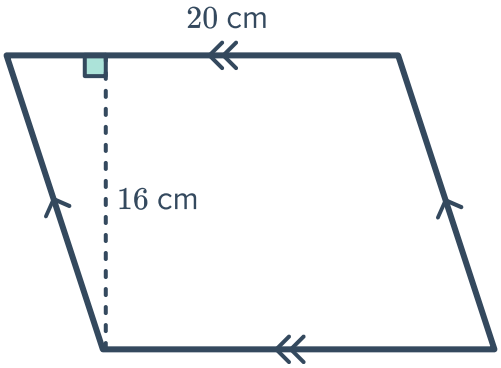
Mixed areas

19 Find the area of the following quadrilaterals:

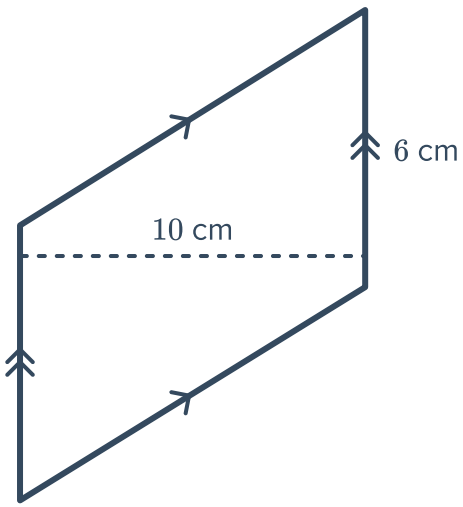
a



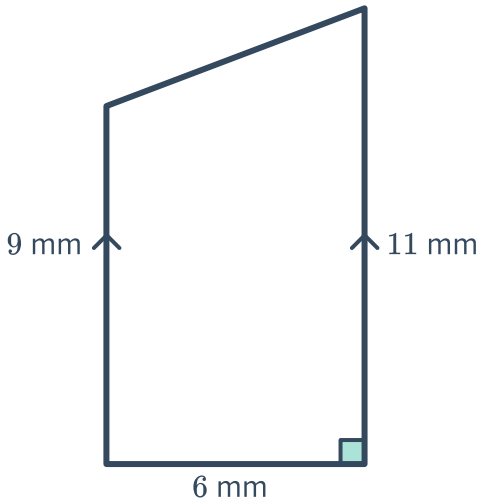
b



c

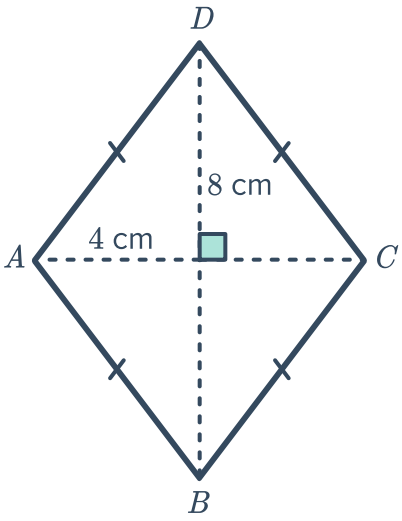


d

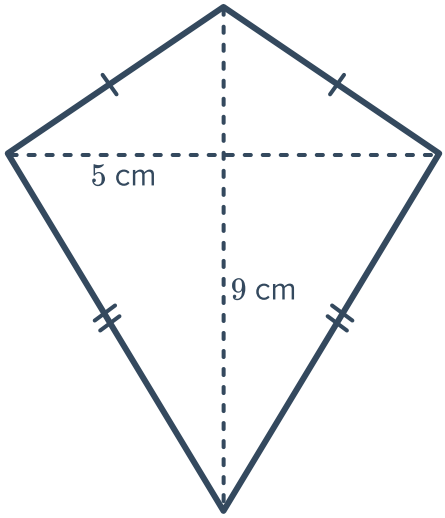


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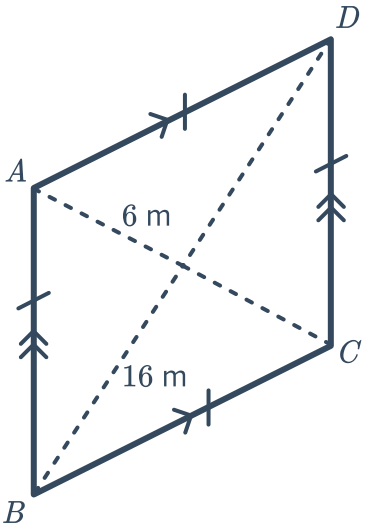
e



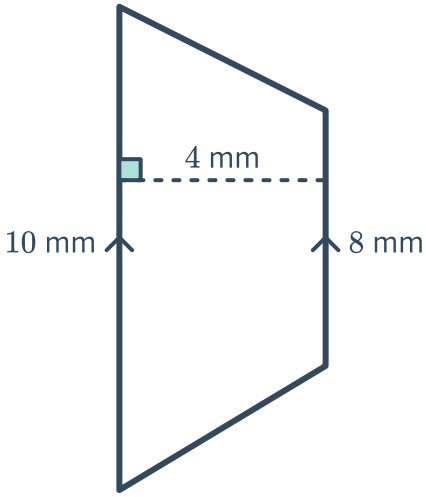
f



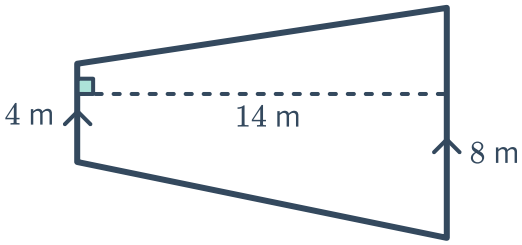
g



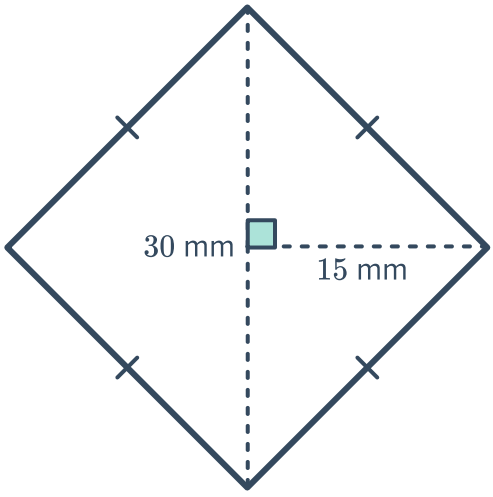
h



i



j

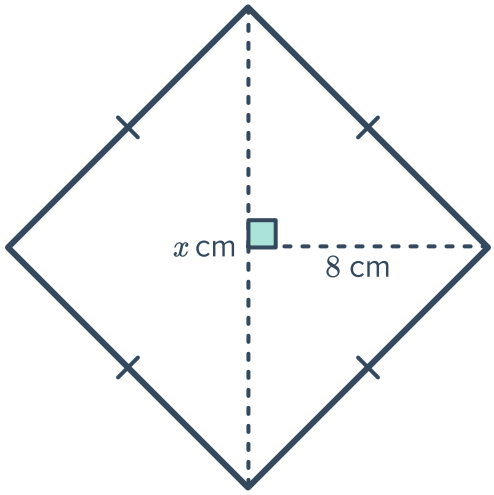


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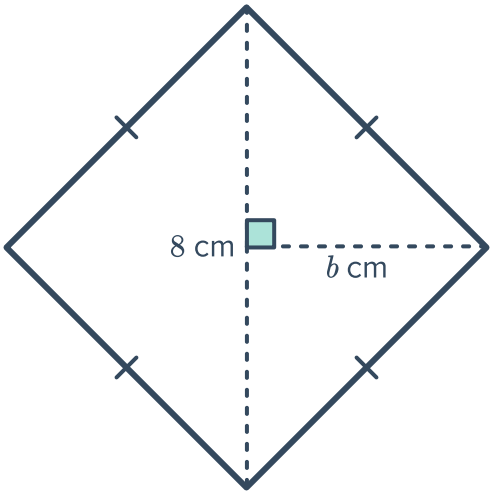
Missing lengths

20 For each of the following rhombuses, find the value of the pronumeral:

a $A = 64\ \text{cm}^2$

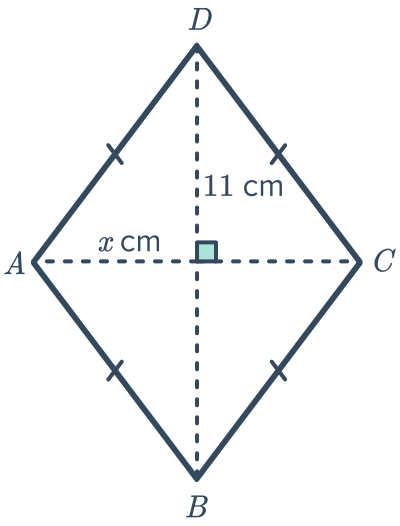


b $A = 128\ \text{cm}^2$



21 Rhombus $ABCD$ has an area of $A = 55\ \text{cm}^2$:

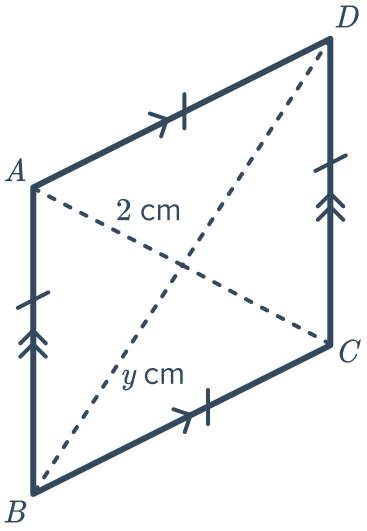
Given the diagonal $BD = 11\ \text{cm}$, and $AC = x\ \text{cm}$, find the value of x.



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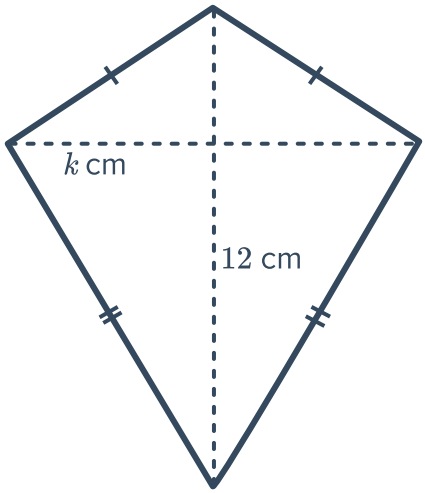
22 Rhombus $ABCD$ has an area of $13\ \text{cm}^2$:

If diagonal $AC = 2$, and diagonal $BD = y$, find the value of y.



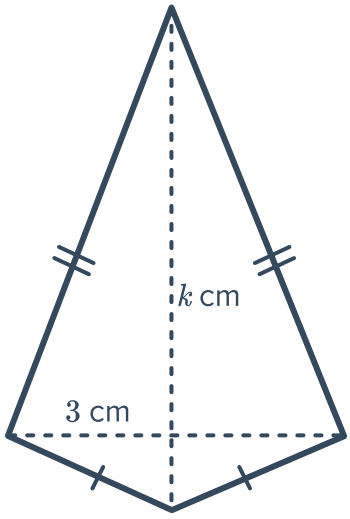
23 The following kite has an area of $48\ \text{cm}^2$.
The length of one of its diagonals is $12\ \text{cm}$:

Find the length of the other diagonal, k.

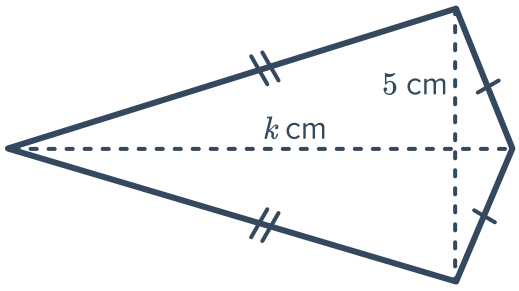


24 For each of the following kites, find the value of k:

a $A = 15\ \text{cm}^2$

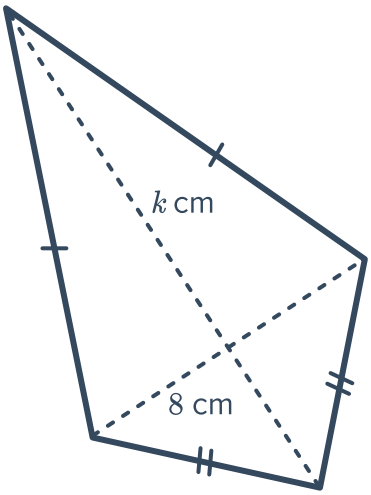


b $A = 22.5\ \text{cm}^2$

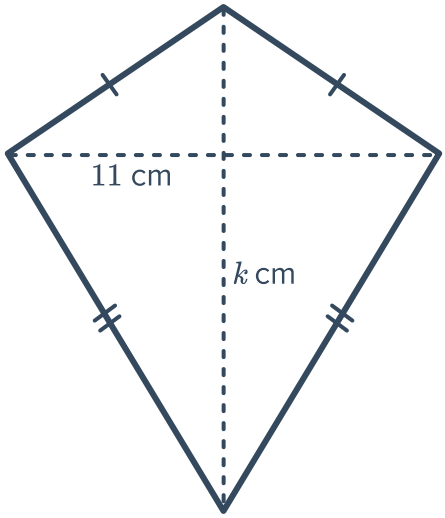


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c $A = 56\ \text{cm}^2$

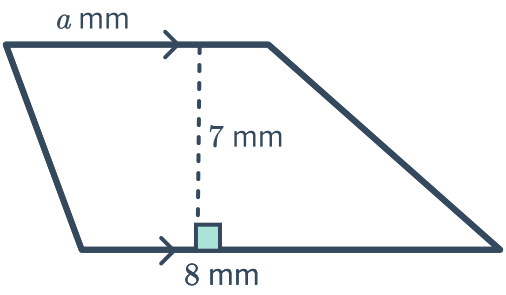


d $A = 137.5\ \text{cm}^2$

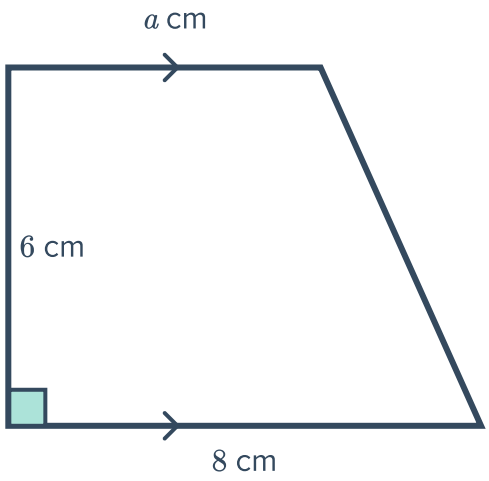


25 For each of the following trapezia, find the value of the pronumeral:

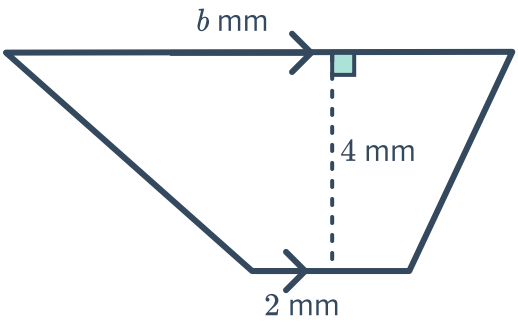
a $A = 42\,\text{mm}^2$



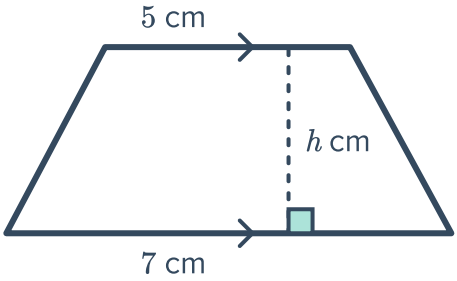
b $A = 36\ \text{cm}^2$



c $A = 20\ \text{m}^2$

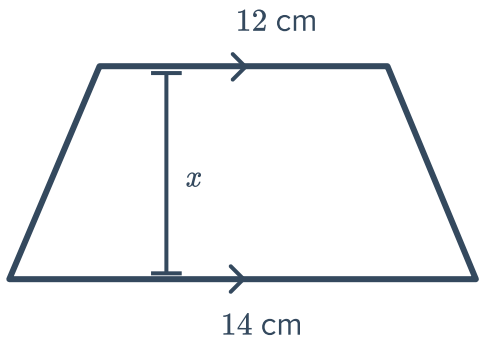


d $A = 24\ \text{cm}^2$



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26 Find the value of x if the area of the trapezium shown is $65\ \text{cm}^2$:



27 Find the perpendicular height, h, of a parallelogram that has an area of $45\ \text{cm}^2$ and a base of $5\ \text{cm}$.

28 Find the base length, b, of a parallelogram that has an area of $216\ \text{mm}^2$ and a perpendicular height of $12\ \text{mm}$.

29 The area of a kite is $640\ \text{cm}^2$ and one of the diagonals is $59\ \text{cm}$. If the length of the other diagonal is $y\ \text{cm}$, find the value of y, rounded to two decimal places.

30 Complete the table of base and height measurements for three parallelograms that all have an area of $24\ \text{m}^2$:

Area (m^2)	Base (m)	Height (m)
24		8
24	12	
24	6	

31 Complete the table og the lengths of diagonal x and diagonal y for three kites that all have an area of $36\ \text{mm}^2$:

Area (mm^2)	Diagonal, x (mm)	Diagonal, y (mm)
36		18
36	24	
36	12	

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